

ORIGINAL ARTICLES

Effects on coronary artery disease of lipid-lowering diet, or diet plus cholestyramine, in the St Thomas' Atherosclerosis Regression Study (STARS)

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To assess the effect of dietary reduction of plasma cholesterol concentrations on coronary atherosclerosis, we set up a randomised, controlled, end-point-blinded trial based on quantitative image analysis of coronary angiograms in patients with angina or past myocardial infarction. Another intervention group received diet and cholestyramine, to determine the effect of a greater reduction in circulating cholesterol concentrations.

90 men with coronary heart disease (CHD), who had a mean (SD) plasma cholesterol of 7.23 (0.77) mmol/l were randomised to receive usual care (U, controls), dietary intervention (D), or diet plus cholestyramine (DC), with angiography at baseline and at 39 (SD 3.5) months. Mean plasma cholesterol during the trial period was 6.93 (U), 6.17 (D), and 5.56 (DC) mmol/l. The proportion of patients who showed overall progression of coronary narrowing was significantly reduced by both interventions (U 46%, D 15%, DC 12%), whereas the proportion who showed an increase in luminal diameter rose significantly (U 4%, D 38%, DC 33%). The mean absolute width of the coronary segments (MAWS) studied decreased by 0.201 mm in controls, increased by 0.003 mm in group D, and increased by 0.103 mm in group DC ($p < 0.05$), with improvement also seen in the minimum width of segments, percentage diameter stenosis, and edge-irregularity index in intervention groups. The change in MAWS was independently and significantly correlated with LDL cholesterol concentration and LDL/HDL cholesterol ratio during the trial period. Both interventions significantly reduced the frequency of total cardiovascular events.

Dietary change alone retarded overall progression and increased overall regression of coronary artery disease, and diet plus cholestyramine was

additionally associated with a net increase in coronary lumen diameter. These findings support the use of a lipid-lowering diet, and if necessary of appropriate drug treatment, in men with CHD who have even mildly raised serum cholesterol concentrations.

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Introduction

Diet is the main environmental determinant of plasma lipid concentrations, and dietary modification is recognised to be the first line of treatment for hyperlipidaemia.¹ We have previously reported angiographic evidence that treatment of hyperlipidaemia by diet and medication retards progression of symptomatic atherosclerosis in femoral arteries.² Several trials have shown that treatment with lipid-lowering drugs may lead to regression of coronary atherosclerosis. That dietary modification may have a similar effect is supported by an uncontrolled trial,³ by a trial in which diet was used as one of five interventions,⁴ and by findings in a diet-treated placebo group.⁵

The St Thomas' Atheroma Regression Study (STARS) was designed to assess the effects of a practicable lipid-lowering diet on the coronary arteries of patients with coronary heart disease (CHD) by computerised image analysis of paired coronary angiograms at an interval of about 3 years. A second intervention group followed the diet and received the lipid-lowering drug cholestyramine,¹ to observe the effects of a greater reduction in plasma cholesterol concentration.

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Patients and methods

Patients

Men referred for coronary angiography to investigate angina pectoris or other findings suggestive of CHD were eligible for entry to the study if they were under 66 years of age and had plasma cholesterol concentrations above 6.0 mmol/l in two consecutive blood samples while on their usual diet. This cut-off value was set to obtain widely generalisable results; in a study of British men with CHD, mean plasma cholesterol was 6.9 mmol/l.⁶ Patients with plasma cholesterol concentrations above 10 mmol/l were excluded to avoid inappropriate randomisation of patients with severe hypercholesterolaemia to a control group, as were patients already treated with lipid-lowering drugs.

Predetermined grounds for exclusion were fasting plasma triglyceride concentrations above 4 mmol/l or remnant hyperlipidaemia (both might be adversely affected by cholestyramine), fasting glucose above 7 mmol/l, cardiac failure, myocardial infarction in the previous 8 weeks, malignancy, other major organ disease, or accelerated hypertension. Patients were also excluded if initial angiographic or clinical findings indicated a need for coronary artery revascularisation or if they had undergone such a procedure. Patients in the study who subsequently required single-vessel or double-vessel bypass grafts or single-vessel angioplasty were included in the analysis but the angiographic appearance of the grafts, grafted arteries, or arteries that had undergone angioplasty were not used for comparison of outcome.

The nature and purpose of the trial were described to patients during the initial admission for angiography and a fasting blood sample was obtained on the morning after admission. Eligible patients were seen again twice for repeat fasting blood samples, and at the second of these visits the trial was again described and written consent obtained. Patients thought unable to understand the implications of the trial design were excluded. The research protocol was approved by the St Thomas' Hospital and Central Oxford ethics committees.

All eligible patients then underwent a short trial of cholestyramine to identify those who responded to and were tolerant of the drug. 8 g cholestyramine twice daily was prescribed for 14 days, after which patients reattended for a fasting blood sample. Patients were included in the study if they agreed to continue cholestyramine at the same dosage for 3 years should they be randomised into the appropriate group and if plasma cholesterol had fallen by more than 0.5 mmol/l on cholestyramine alone. At this visit eligible patients were randomly assigned to one of three groups: a lipid-lowering diet (group D); diet plus cholestyramine (DC); and usual care only (U).

Interventions

In the lipid-lowering diet, which has been used in the St Thomas' Hospital lipid clinic since 1982, total fat intake was reduced to 27% of dietary energy, saturated fatty acid content to 8–10% of dietary energy, and dietary cholesterol to 100 mg/1000 kcal; omega-6 and omega-3 polyunsaturated fatty acids were increased to 8% of dietary energy, and plant-derived soluble fibre (chiefly pectin) intake was increased to the equivalent of 3.6 g polygalacturonate/1000 kcal. Intake of alcohol was permitted at the patient's habitual level. Patients in groups D and DC were instructed by a dietitian after a detailed history was obtained as a guide to energy requirements; choice of food was adapted to individual preferences. For patients with a body mass index (BMI) below 25 kg/m² an isocaloric diet was prescribed; overweight patients were prescribed a diet that contained 1000–1200 kcal daily to achieve a BMI of 25 kg/m². Further dietetic counselling and, if requested, suitable foodstuffs, were given to participants in the diet or diet plus cholestyramine groups who did not achieve or maintain serum cholesterol reductions of 15% or 25%, respectively. At each follow-up visit the clinician inquired about dietary compliance and provided encouragement.

Group DC received similar dietetic advice together with 8 g cholestyramine twice daily with meals. Unused sachets were counted at follow-up visits; mean (SD) daily intake was assessed at 14 (2.4) g.

TABLE 1—CLINICAL AND DEMOGRAPHIC CHARACTERISTICS

Characteristic	Study completed			Did not complete (n=16)
	U (n=24)	D (n=26)	DC (n=24)	
Age (yr)	53.9 (1.6)	48.9 (1.6)	49.6 (1.5)	50.0 (1.9)
Weight (kg)	77.5 (2.1)	79.5 (2.0)	82.5 (2.3)	79.4 (2.4)
Body mass index (kg/m ²)	26.5 (0.6)	26.0 (0.4)	26.0 (0.5)	26.6 (0.9)
Ex-smokers (%)	66.7	65.4	62.5	56.2
Current smokers (%)	29.2	26.9	20.8	37.5
Regular alcohol (%)	83.3	84.6	75.0	75.0
Units alcohol/week	15 (3)	10 (2)	12 (2)	9 (2)
Type A personality (%)	62.5	65.4	70.8	50.0
Social class I or II (%)	54.2	50.0	62.5	43.7
History of hypertension (%)	8.3	7.7	8.3	6.2
Family history of early CHD (%)	45.8	50.0	41.6	37.5
Previous MI (%)	41.7	46.1	54.2	56.2
Angina at entry (%)	75.0	69.2	79.2	62.5

Mean (SEM) values shown for continuous variables U = usual care, D = diet, DC = diet plus cholestyramine groups; CHD = coronary heart disease, MI = myocardial infarction

The control group received, in common with both intervention groups, cardiological supervision and treatment, repeated counselling against smoking, and antihypertensive treatment if appropriate. All participants were advised about a suitable level of daily exercise. Patients in the control group with a BMI above 25 kg/m² were advised to lose weight but did not receive formal dietary counselling.

Follow-up

Patients were seen at intervals of about 3 months for clinical assessment, fasting venous blood samples, analysis and encouragement of compliance with treatment, and further dietary counselling where appropriate. Supine blood pressure was measured with a standard sphygmomanometer after 5 min and mean arterial blood pressure calculated as diastolic pressure (at Korotkov phase 5) plus $\frac{1}{3}$ pulse pressure. Angina was assessed as number of episodes per week, duration in min, severity on a scale of 1 to 7 (7 worst), and by Canadian Heart Association functional class.⁷ Changes in medication were recorded. Patients underwent repeat angiography approximately 3 years after randomisation (mean [SD] 39 [3.5] months).

Coronary angiography and image analysis

We used the Judkins percutaneous femoral artery technique for coronary angiography, without routine use of vasodilators, and with almost exclusive use of 'Cordis' 8 F catheters (Miami, Florida, USA). 4 views of the right and 7 of the left coronary artery were routinely recorded on 35 mm cine-film at 60 frames/s by use of a caesium iodide intensifier with a 1 cm² grid for correction of pincushion distortion. Viewing angles, table height, and the sequence of views were recorded and reproduced as exactly as possible, by the same cardiologist, at repeat angiography.

Coronary artery segments 1–3, 5–8, and 11–13⁸ were analysed if there was a clear unobstructed view, and the segment was oriented in the image plane and clearly seen in at least 2 different views. Paired, coded films were compared on a twin-channel projector and paired frames of the same view at end-diastole were photographed; the photographs were coded, and marked to identify the segments for image analysis. Coding was done by an investigator who was aware of the film sequence but blinded to the therapeutic group; comparison of the coded films and image analysis were done by investigators blinded to the sequence and therapeutic group of the angiograms and photographs.

Selected cine-frames were digitised at high resolution (1024 × 1024 pixels, 8-bit grey scale) by use of a 'Videk Megaplus' CCD array camera (Eastman Kodak, Rochester, New York, USA), and stored on optical disc. Usually 4 projections were stored to ensure adequate views of as many arterial segments as possible. Edges of segments and of the catheter were delineated by computer

TABLE II—PLASMA LIPIDS AND LIPOPROTEINS AT BASELINE AND DURING TRIAL

—	U (n = 24)			D (n = 26)			DC (n = 24)		
	Baseline	In-trial	p	Baseline	In-trial	p	Baseline	In-trial	p
Cholesterol (mmol/l)	7.07 (0.03)	6.93 (0.20)	0.15	7.19 (0.08)	6.17 (0.09)	<0.001	7.44 (0.20)	5.56 (0.16)	<0.001
LDL cholesterol (mmol/l)	4.82 (0.18)	4.67 (0.19)	0.24	5.00 (0.10)	4.19 (0.10)	<0.001	5.26 (0.22)	3.37 (0.19)	<0.001
HDL cholesterol (mmol/l)	1.22 (0.04)	1.21 (0.06)	0.62	1.14 (0.04)	1.14 (0.03)	0.82	1.24 (0.05)	1.19 (0.03)	0.29
Triglyceride (mmol/l)	2.32 (0.24)	2.35 (0.20)	0.86	2.31 (0.21)	1.85 (0.20)	<0.005	2.20 (0.18)	2.21 (0.22)	0.97
Cholesterol/HDL cholesterol	5.92 (0.22)	6.08 (0.34)	0.50	6.52 (0.25)	5.50 (0.18)	<0.001	6.21 (0.31)	4.71 (0.17)	<0.001
LDL/HDL cholesterol	4.05 (0.20)	4.10 (0.25)	0.75	4.54 (0.19)	3.74 (0.14)	<0.001	4.39 (0.27)	2.87 (0.19)	<0.001

Mean (SEM) values shown; within-group comparisons by paired *t*-test. U = usual care, D = diet, DC = diet plus cholestyramine groups

according to the methods of Brunt and colleagues,⁹⁻¹¹ and for each paired projection this process was usually repeated in three adjacent frames to improve precision; all detected edge-points were stored on disc. From these stored edge-points, corrected for pincushion distortion from the intensifier system, the mean width along each segment, expressed in pixels, was computed. Mean absolute width of the segment (MAWS) and minimum absolute width of each segment (MinAWS), expressed in mm, were calculated from catheter dimensions obtained by micrometry.

Measures of outcome

Change in MAWS was the principal end-point measure, and an indication of overall change in coronary atherosclerosis for each patient was given as an overall MAWS change, averaged from all changes obtained from the measurable segments. The definition of disease progression and regression for patients and for individual segments took into account the imprecision (2 SD, calculated from the variability of repeated analysis of arterial segments) of measurement of the change in MAWS; overall and segmental progression was thus defined as a change exceeding -0.17 mm and -0.24 mm, respectively, and regression as change exceeding +0.17 mm and +0.24 mm diameter, respectively. Secondary angiographic end-points were the change in MinAWS (mm), percentage diameter stenosis,¹² and the edge-irregularity index (%).¹³ Because totally occluded segments could not undergo edge-detection, and overall assessment would be seriously incomplete without taking them into account, we also defined a "functional" segmental width which would be zero if blood did not reach that segment because of occlusion of a more proximal segment. These and other approaches to quantitative coronary angiography have been reviewed.¹⁴

Clinical end-points recorded were deaths (by cause), non-fatal myocardial infarction, coronary artery bypass surgery or angioplasty during the trial, stroke, frequency and degree of angina, and change in anti-anginal medication.

Laboratory investigations included fasting (12 h) plasma lipid and lipoprotein concentrations at each clinic visit, with plasma separated immediately and analysis started the same day. Cholesterol and triglyceride concentrations were measured enzymatically, with mean interassay coefficients of variation of 2.8% at 3.0 mmol/l and 3.4% at 7.6 mmol/l for cholesterol; and 3.0% at 1.8 mmol/l and 2.8% at 3.0 mmol/l for triglyceride. High density lipoprotein (HDL) was isolated by precipitation of very low density (VLDL) and low density lipoproteins (LDL) with heparin-manganese chloride; excess manganese was chelated by sodium-edetic acid before cholesterol assay. LDL cholesterol was derived by the Friedewald formula. Haematocrit, liver and thyroid function, and glucose and creatinine concentrations were measured at baseline and 6-monthly thereafter.

TABLE III—BASELINE ANGIOGRAPHIC DATA

—	U (n = 24)	D (n = 26)	DC (n = 24)
Segments analysed	157	169	163
Diameter stenosis (%)	18.6 (1.9)	23.3 (2.4)	23.4 (2.0)
EII (%)	8.0 (0.7)	9.8 (0.9)	9.9 (0.8)
Total occluded segments	4	7	4
Patients with occlusions	4	7	3
Ejection fraction	0.54 (0.02)	0.54 (0.02)	0.56 (0.03)

Mean (SEM) values shown for continuous variables. EII = edge-irregularity index; U = usual care, D = diet, and DC = diet plus cholestyramine groups

Statistical methods

Changes in plasma lipids and lipoproteins, weight, blood pressure, and smoking were examined by the paired Student's *t*-test. One-way analysis of variance and the unpaired *t*-test were used to compare group differences in the changes of MAWS, MinAWS, percentage diameter stenosis, and edge-irregularity index by patient (overall score) and by segment. The influence of group differences in baseline variables on differences of changes in MAWS was studied by analysis of covariance. Differences in the proportion of patients and segments that showed progression or regression were compared by the χ^2 and Fisher's exact tests. Group differences in the proportion of patients with definite clinical end-points were also compared by Fisher's exact test. Baseline, in-trial, and changes in recorded variables associated with changes in MAWS were identified by stepwise multiple linear regression. Finally, a multivariable linear equation was derived to predict MAWS changes by model-fitting analysis of variables that were not co-correlated and achieved a significance of *p* < 0.1 in univariate regression.

Results

90 patients were randomised, of whom 74 completed the study with paired angiograms suitable for analysis at a mean (SD) interval of 39 (3.5) months. 4 patients died suddenly; 10 withdrew voluntarily after a mean of 6 months because of unwillingness to undergo a second angiogram (2), intolerance to cholestyramine (3), emigration (2), or neuropsychiatric disturbance (2); 1 had a minor stroke after 24 months and was withdrawn; and 2 had only one angiogram available for analysis.

Table I shows the demographic and clinical characteristics of the patients. There were no statistically significant differences in the characteristics shown between patients who completed or did not complete the study, nor between the treatment groups.

Weight and blood pressure measurements were similar among the treatment groups before and during the study. Cigarette use tended to decrease but within-group and between-group differences were not statistically significant.

Plasma lipids and lipoproteins

Table II shows plasma lipid and lipoprotein concentrations and lipoprotein ratios by treatment group. There were no significant group differences at outset with

TABLE IV—ANGIOGRAPHIC CHANGES BY PATIENT

Changes in:	U (n = 24)	D (n = 26)	DC (n = 24)	p for trend*
MAWS (mm)	-0.201 (0.062)	0.003 (0.087)†	0.103 (0.051)§	0.012
MinAWS (mm)	-0.232 (0.068)	0.030 (0.086)‡	0.117 (0.051)§	0.003
% DS	5.8 (1.8)	-1.1 (3.7)	-1.9 (1.1)¶	0.077
EII (%)	2.0 (0.7)	0.0 (1.1)	-0.8 (0.5)¶	0.066

*By analysis of variance. †*p* = 0.06, ‡*p* < 0.05, §*p* < 0.001, ¶*p* < 0.01 vs usual care group. MAWS = mean absolute width of segment, MinAWS = minimum absolute width of segment, % DS = percentage diameter stenosis; EII = edge-irregularity index; and U = usual care, D = diet, and DC = diet plus cholestyramine groups.

TABLE V—ANGIOGRAPHIC CHANGES BY SEGMENT

Changes observed	U	D	DC	p for trend*
<i>All segments (n=489)</i>	<i>(n=157)</i>	<i>(n=169)</i>	<i>(n=163)</i>	
MAWS (mm)	-0.131 (0.118)	0.011 (0.118)†	0.076 (0.074)‡	0.002
MinAWS (mm)	-0.161 (0.111)	0.034 (0.116)‡	0.086 (0.084)†	<0.0001
% DS	5.6 (3.6)	-0.5 (4.0)§	-1.5 (2.7)†	<0.0001
EII (%)	1.9 (1.3)	0.0 (1.4)	-0.6 (0.9)†	0.001
<i>Patent segments (n=474)</i>	<i>(n=153)</i>	<i>(n=162)</i>	<i>(n=159)</i>	
MAWS (mm)	-0.135 (0.120)	-0.023 (0.110)‡†	0.070 (0.072)‡	0.002
MinAWS (mm)	-0.165 (0.112)	0.006 (0.111)*	0.082 (0.084)†	<0.0001
% DS	5.8 (3.6)	1.1 (3.30)	-1.0 (2.4)‡	0.001
EII (%)	1.9 (1.3)	0.5 (1.2)	-0.5 (0.8)‡	0.001
<i><15% BS (n=208)</i>	<i>(n=73)</i>	<i>(n=72)</i>	<i>(n=63)</i>	
MAWS (mm)	-0.130 (0.142)	0.062 (0.082)	0.055 (0.086)**	0.055
MinAWS (mm)	-0.198 (0.134)	-0.011 (0.099)**	0.020 (0.090)	0.044
% DS	8.8 (4.3)	4.4 (1.9)	2.5 (1.5)	0.035
EII (%)	2.8 (1.7)	0.8 (0.6)**	0.5 (0.6)	0.033
<i>15%–50% BS (n=243)</i>	<i>(n=75)</i>	<i>(n=84)</i>	<i>(n=84)</i>	
MAWS (mm)	-0.099 (0.082)	-0.085 (0.125)	0.094 (0.062)*	0.016
MinAWS (mm)	-0.122 (0.086)	0.012 (0.121)	0.105 (0.075)‡	0.014
% DS	2.3 (2.5)	-1.2 (4.1)	-1.2 (2.4)	0.269
EII (%)	1.0 (0.9)	0.5 (1.5)	-0.3 (0.8)	0.333
<i>>50% BS (n=38)</i>	<i>(n=9)</i>	<i>(n=13)</i>	<i>(n=16)</i>	
MAWS (mm)	-0.406 (0.161)	0.345 (0.207)**	0.066 (0.084)	0.098
MinAWS (mm)	-0.187 (0.084)	0.423 (0.157)	0.249 (0.098)	0.075
% DS	7.4 (3.9)	-23.3 (7.9)	-18.4 (4.7)*	0.05
EII (%)	2.0 (0.7)	-7.6 (2.9)	-6.7 (1.6)**	0.078

Mean (SEM) values shown. BS = baseline stenosis, other abbreviations as for table IV.
*By analysis of variance
†p<0.0001, ‡p<0.001, §p<0.005, ||p<0.01, |||p<0.05, **p=0.06, and ††p=0.08 vs usual care group by unpaired *t*-test.

respect to plasma cholesterol, LDL cholesterol, HDL cholesterol, triglyceride, cholesterol/HDL cholesterol ratio, or HDL/LDL cholesterol ratio. In group U the mean values were very stable during the study. In group D there were falls in plasma cholesterol (14.2%), LDL cholesterol (16.2%), triglyceride (20%), plasma cholesterol/HDL cholesterol ratio (15.6%), and LDL/HDL ratio (17.6%). In group DC the falls in cholesterol (25.3%), LDL cholesterol (35.7%), cholesterol/HDL cholesterol ratio (24.1%), and LDL/HDL cholesterol ratio (34.6%) were greater than those observed in group D, but there was no significant reduction in triglyceride. Plasma HDL cholesterol concentration did not alter significantly in groups D or DC.

Angiographic findings

Paired measurements were made on 489 segments, with a mean (SD) of 6.4 (2.0) segments analysed per patient (no significant differences between groups). Baseline angiographic data by treatment group are given in table III. Because the three patients in group U who died had severe CHD at entry, mean baseline values for diameter stenosis and edge-irregularity index in the surviving patients were lowest in this group; however, between-group differences are not statistically significant. Total numbers of occluded segments, patients with occlusions, and mean ejection fractions, were similar between groups.

Table IV shows the mean within-patient changes in quantitative estimates of coronary artery disease by treatment group. MAWS fell by 0.201 mm in group U, but increased by 0.003 mm in group D, and increased by 0.103 mm in group DC (*p*<0.012 for trend); 95% confidence intervals for mean differences between groups were -0.424, 0.016 mm (U vs D) and -0.470, -0.139 mm (U vs DC). Similar trends were seen with other measures of angiographic change (table IV). Group differences in outcome measures were not significantly affected by

differences in baseline variables. Assessment of the distribution of results, in which the few extreme values recorded were excluded, showed similar differences between treatment groups but the reduced variation led to smaller *p* values such that group differences were significant for all outcome measures.

Overall progression was observed in 11 patients (46%) in group U, 4 (15%) in group D, and 3 (12%) in group DC (*p*<0.02 for trend, χ^2 test). Regression of atherosclerosis occurred in 1 patient (4%) in group U, 10 patients (38%) in group D, and 8 (33%) in group DC (*p*<0.02 for trend). Ejection fraction did not alter significantly in any of the groups. The mean interval (SD) between paired angiograms was not significantly different between groups: U 40.0 (4.5), D 38.1 (3.5), and DC 38.9 (2.8) months.

The effects of treatment by segment are shown in table v, grouped for all segments, all patent segments, and by baseline severity of disease (minimal, <15% diameter stenosis; moderate, 15–50% diameter stenosis; and severe, >50% stenosis). All angiographic criteria consistently worsened in group U and consistently improved in groups D and DC for all segments and for all segments patent at outset, with significant trends across all treatment groups. Segments with minimal, moderate, and severe baseline stenosis all showed favourable trends in both intervention groups, but deterioration in the usual care group (table v).

TABLE VI—CARDIOVASCULAR EVENTS

—	U	D	DC
Patients at risk*	28	27	26
Deaths (%)	3 (11)	1 (4)	0
Myocardial infarction (%)	2 (7)	1 (4)	1 (4)
Coronary surgery (%)	3 (11)	1 (4)	0
Angioplasty (%)	1 (4)	0	0
Stroke (%)	1 (4)	0	0
Total cardiac events (%)	10 (36)	3 (11)†	1 (4)‡

*Only excludes patients who withdrew
†p<0.05, ‡p<0.01 vs usual care group by Fisher's exact test

TABLE VII—CORRELATIONS BETWEEN CHANGE IN MAWS AND CHANGES FROM BASELINE OR MEAN IN-TRIAL PLASMA LIPID AND LIPOPROTEIN

—	Change from baseline		Mean in-trial concentration	
	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>
Cholesterol	0.25	0.03	−0.32	0.004
LDL cholesterol	0.27	0.02	−0.40	<0.001
HDL cholesterol	−0.07	0.53	0.23	0.048
Triglyceride	0.08	0.48	0.05	0.64
Cholesterol/HDL cholesterol	0.19	0.09	−0.36	0.001
LDL/HDL cholesterol	0.27	0.02	−0.42	<0.001

Pooled data, *n* = 74

The most striking changes in diameter stenosis were seen in the worst affected segments.

Among controls 34% of segments showed progression, significantly more (*p* < 0.01 for trend) than in both intervention groups (group D, 22%; group DC, 17%). More segments showed regression in groups D and DC than in controls, but the trend was not significant. 5 segments became occluded in group U, 3 in group D, and 0 in group DC (*p* < 0.09 *vs* UC); 1 occluded segment recanalised during the study, in group DC.

Clinical events

14 patients had clinical cardiovascular events as shown in table VI, 10 of whom were controls (*p* < 0.05 *vs* diet, *p* < 0.01 *vs* diet plus cholestyramine). All deaths were sudden and were not caused by trauma or non-cardiovascular disease. Angina symptoms were not significantly changed among controls, whereas both intervention groups showed significant improvements (*p* < 0.01) in severity of angina and CHA functional score. The change in weekly frequency of angina was correlated with the change in MAWS (*r* = 0.31, *p* = 0.008).

During the study β-blockers were used by 35% of patients, calcium antagonists by 34%, long-acting oral nitrates by 24%, diuretics by 19%, aspirin by 18%, and dipyridamole by 13%. The proportion of patients taking these drugs did not differ significantly among treatment groups. 10 controls (42%) required increased anti-anginal treatment compared with 5 (19%) in group D and 6 (25%) in group DC. There was no significant correlation between change in medication and change in MAWS.

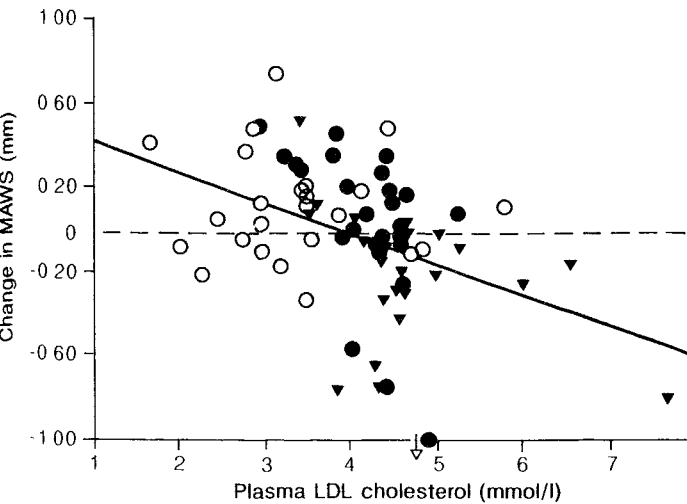


Fig 1—Correlation of mean absolute width of segment (by patient) and mean plasma LDL cholesterol concentration during trial.
 Δ MAWS = 0.58 − 0.15 × LDL, *r* = −0.40; *p* < 0.001.
▼ = control, ● = diet; ○ = diet plus cholestyramine.

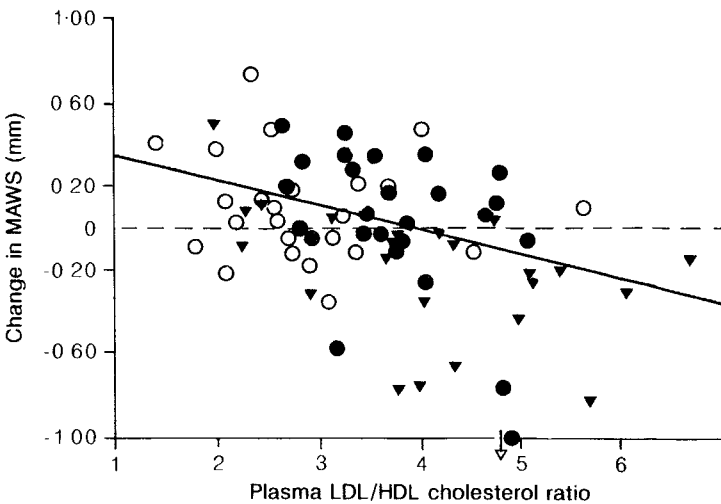


Fig 2—Correlation of mean absolute width of segment (by patient) and in-trial LDL/HDL cholesterol ratio.
 Δ MAWS = 0.48 − 0.14 × LDL/HDL, *r* = −0.42, *p* < 0.001.
Symbols as for fig 1.

Correlations with change in MAWS

Of baseline variables, only percentage diameter stenosis was significantly correlated with change in MAWS (*r* = 0.28, *p* = 0.016; by patient); it was also associated with change in MinAWS (*r* = 0.36, *p* < 0.01), change in percentage diameter stenosis (*r* = −0.55, *p* < 0.001), and change in edge-irregularity index (*r* = −0.50, *p* < 0.001). Mean lipid and lipoprotein concentrations during the trial were more strongly correlated with the change in MAWS than were the changes from baseline (table VII); strongest correlations were seen with mean LDL cholesterol (fig 1) and LDL/HDL cholesterol ratio (fig 2). Associations were also observed between the change in MAWS and in mean arterial blood pressure (*r* = 0.24, *p* = 0.04) and treatment group (*r* = −0.34, *p* = 0.003), but the change in MAWS was independent of treatment group if the in-trial LDL cholesterol concentration or LDL/HDL cholesterol ratio were taken into account. No significant correlations were found with age, BMI, current smoking or change in smoking habits, alcohol consumption, personality type, systolic or diastolic blood pressures, or use of anti-anginal drugs.

The best multilinear regression model to estimate change in MAWS included change in mean arterial blood pressure (MBP) and the mean LDL/HDL cholesterol ratio during the trial (Δ MAWS = 0.015 × Δ MBP − 0.145 × LDL/HDL + 0.47; *r* = 0.50; *p* < 0.0001).

Discussion

As in other trials of lipid lowering in atherosclerotic femoral^{2,15} and coronary^{3,4,16-19} artery disease, the STARS trial used as an end-point arteriographic assessment of the arterial lumen, which may not necessarily equate with progression and regression of atherosclerosis. However, comparison with the effects of lipid-lowering studies in animals²⁰⁻²² suggests that such trials in man at least partly reflect favourable effects on the natural history of atherosclerosis. Lysis of arterial thrombus might contribute to luminal widening, but persistent angiographic changes at an average interval of 39 months would be unlikely; we have no reason to suspect that controls had more thrombi at presentation than the intervention groups. Experimental hypercholesterolaemia impairs endothelium-mediated vasodilator responses,^{23,24} which in animals are partly restored by a low-fat diet.²⁵ Such an effect could contribute to luminal widening seen in “regression” studies. The

consistent associations observed between arteriographic improvement and clinical end-points also suggest that such measurements are related to cardiac prognosis.^{17,18} However, macroscopic evidence of regression of atherosclerosis in man is limited to necropsy series which show fewer arterial lesions during wartime nutritional deprivation²⁶ or, less consistently, among patients who die from cancer.²⁷

In this study of men with CHD, coronary atherosclerosis, and mild-to-moderate hypercholesterolaemia, almost half (46%) the controls, who received usual medical treatment, showed progressive coronary narrowing and only 1 (4%) showed luminal widening. By contrast, subjects advised to follow a lipid-lowering diet, and those on diet and cholestyramine, showed a 66% fall in the incidence of progression, a 10-fold increase in the frequency of luminal widening, and fewer cardiac events; patients in the diet plus cholestyramine group also showed widening of mean coronary diameter and of stenoses, compared with controls.

Other investigators have studied the effects of lipid-lowering diets on coronary disease. In the Leiden trial,³ which was not controlled, a vegetarian diet with a very high ratio of polyunsaturated fat to saturated fat was employed and favourable coronary angiographic changes correlated with the fall in serum cholesterol and the rise in HDL cholesterol. In the Lifestyle Heart Trial,⁴ a restrictive diet (<10% energy from fat) was employed as one of five interventions; a favourable effect on coronary stenoses was observed, but the investigators did not claim that this regimen was practicable. The lipid-lowering diet used in our trial was readily accepted and has been successfully used for a decade.^{28,29} With careful dietary counselling and motivation, most patients in the STARS study complied quite well with the diet for over 3 years and it thus represents the first randomised, controlled trial of moderate dietary change as a unifactorial intervention in patients with coronary atherosclerosis. A second intervention group received dietary advice plus cholestyramine to investigate the effect of a further reduction of LDL cholesterol. Cholestyramine reduces the incidence³⁰ and progression³¹ of CHD, has a good safety record, is well tolerated in modest dosage, and is still recommended for drug treatment of hypercholesterolaemia.¹ We found diet plus cholestyramine to be more effective than diet alone, except in patients with >50% stenosis at outset, and led to a greater degree of regression of coronary artery disease. Other trials based on coronary angiography have also shown a consistently favourable effect of more intensive lipid-lowering treatment on coronary artery stenoses.^{4,16-19}

Analysis of up to 10 coronary artery segments, provided that they were clearly seen, allowed examination of relatively normal coronary arteries with less than 15% stenosis at outset, of segments affected by moderate stenoses, and of severely stenosed segments. As with other lipid-lowering trials,^{16,18} the most favourable effects tended to be seen in the segments or stenoses which were most severely affected. The plaques most likely to cause acute coronary events are lipid-rich lesions of moderate size associated on average with 30-40% stenosis;^{32,33} we observed significant regression or reduction in progression in such segments. Results are presented by patient (a measure of overall trend¹⁸ which is comparable to visual overall assessment of angiograms¹⁶) and by segment, which greatly increases the number of observations and permits separate assessment of effects on lesions of different initial severity. Segment-specific analysis is based on the absolute width of coronary segments³ rather than percentage stenosis alone, because the latter

measurement assumes normality of the adjacent segment. When segment diameters were treated as continuous variables rather than categorical ones (progression or regression alone), change in diameter was found to be related to mean LDL cholesterol concentration during the trial. The correlation was modest ($r = -0.40$), as found in other trials,^{16,18,19} but highly significant ($p < 0.001$). The lower the LDL cholesterol concentration, the more striking the regression of coronary atherosclerosis. Accordingly, the lower LDL cholesterol concentrations achieved by diet plus cholestyramine, compared with diet alone, were associated with a greater increase in coronary artery diameter. However, patient-specific analysis showed no difference in the incidence of progression and regression between the two interventions—which may reflect a greater sensitivity of segment-specific analysis or a greater degree of improvement in vessel diameter in those patients in the DC group who also showed an overall improvement.

Most, though not all, workers believe cholesterol lowering to be justified in patients with CHD, and a recent meta-analysis of lipid-lowering trials for secondary prevention indicates a substantial and highly significant reduction of further coronary events.³⁴ Studies of atherosclerosis regression, such as STARS, complement trials based on clinical end-points and show that improvement in angiographic appearance, or arrest of progression, may be expected in many patients who receive dietary advice and/or lipid-lowering drugs. Although the changes achieved in coronary diameter or in percentage stenosis are small the haemodynamic effects, particularly for tight stenoses, may be more substantial; moreover, trends seen during the 1-5 years' follow-up of most trials may be continued with more prolonged therapy. Although several regression trials have shown improved angiographic appearances, these have been achieved at widely different mean LDL cholesterol concentrations of 2.46 mmol/l¹⁶ to 4.45 mmol/l.¹⁹ In this study, the regression line in fig 1 suggests a tendency to improvement when LDL cholesterol was below 3.0-3.5 mmol/l. However, the results achieved at various mean cholesterol concentrations in various trials might also indicate that the extent of reduction in LDL cholesterol may be as important as the absolute concentration attained. In this study we observed a closer correlation between mean in-trial LDL cholesterol concentrations and change in MAWS than between the fall in LDL cholesterol and change in MAWS (table VII).

The findings of the STARS trial show that lipid-lowering dietary intervention is indicated in men with symptomatic coronary atherosclerosis and plasma cholesterol concentrations above 6 mmol/l, provided that their prognosis is not severely limited by myocardial damage or other disease; this benefit may be increased by lipid-lowering drugs. We believe that there is growing evidence to support the use of lipid-lowering intervention, first by diet but with drug treatment added if necessary, in men and women with CHD, irrespective of plasma cholesterol concentration. Because of the high absolute risk of further CHD in such patients, and the fact that they are already under medical supervision, secondary prevention by risk factor reduction is a particularly cost-effective approach to the management of CHD.³⁴

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Mechanism of grass-pollen-induced asthma

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Many asthmatics are sensitive to rye-grass pollen, but pollen grains are too large to penetrate the lower airways. Our aim was to investigate the mechanism by which rye-grass pollen causes asthma.

A major allergen of rye-grass pollen, *Lo1 pIX*, is located in intracellular starch granules within pollen grains. In-vitro tests showed that pollen grains are ruptured in rainwater by osmotic shock, each grain releasing about 700 starch granules into the environment. These granules are small enough to enter the airways (< 3 µm in diameter). The starch granules were present in atmospheric samples taken during the pollen season, and showed a 50-fold increase in atmospheric concentration on days following rainfall. Isolated granules elicited IgE-mediated responses in asthmatic patients, and 4 patients with rainfall-associated asthma who

underwent an inhalation challenge test had striking bronchial constriction after exposure to starch granules.

Starch granules released from rye-grass pollen seem to be capable of causing asthma.

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Introduction

Epidemics of asthma have been associated with thunderstorms and related atmospheric events.¹ On two

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